D 30564 **(Pages : 3)** **Name.....**

Reg. No.....

FIFTH SEMESTER (CBCSS–UG) DEGREE EXAMINATION
NOVEMBER 2022

Mathematics

MTS 5D 01—APPLIED CALCULUS

(2019 Admission only)

Time : Two Hours Maximum : 60 Marks

Section A

All questions can be answered.
Each question carries 2 *marks.*
Ceiling is 20.

1. Find $f\left(\frac{-1}{2}\right)$ and $f(2) = \frac{1}{x-1}, x < 1$
$= 3x^2 + 1, x \geq 1.$

2. Find the domain and range of $g(t) = \sqrt{t-2}$.

3. Find all points of intersection of the graphs of $f(x) = 3x + 2$ and $g(x) = x^2$.

4. Find the equation of the line that passes through the point (5,1) with slope $\frac{1}{2}$.

5. Show that the polynomial $p(x) = 3x^3 - x + 5$ is continuous at $x = 1$.

6. If $s(t) = t^2 - 2t + 6$ for $0 \leq t \leq 2$ is the position of a particle moving along a straight line at time t, find velocity.

7. Solve for x if $\log_4 x = \frac{1}{2}$.

8. Differentiate $\frac{1}{\sqrt{5} - 3x}$.

Turn over

9. Find the intervals of increase of the function $f(x) = x^2 - 2x + 3$.

10. Evaluate $5^{\frac{1}{3}} 2^{\frac{1}{3}}$.

11. Differentiate $f(x) = 5^{2x-3}$.

12. Find $f(x-1)$, if $f(x) = 3x^2 + \frac{1}{x} + 5$.

Section B

All questions can be answered.
Each question carries 5 *marks.*
Ceiling is 30.

13. Find $\lim_{x \to 1} \frac{-x^3 + 2x + 1}{x - 3}$.

14. Differentiate the quotient $\frac{x^2 - 5x + 7}{2x}$.

15. Find the fifth derivative of $f(x) = 4x^3 + 5x^2 + 6x - 1$.

16. Find $\int 8x\left(4x^2 - 3\right)^5 dx$.

17. Find all real numbers x that satisfy the equation $2^{3-x} = 4^x$.

18. Find all points on the graph of the function $f(x) = (x+1)(x^2 - x - 2)$ where the tangent line is horizontal.

19. Find the slope and y-intercept of the line $3y + 2x = 6$ and draw the graph.

Section C

Answer any **one** *question.*
Each question carries 10 marks.

20. Graph the function $f(x) = 2x$, if $0 \leq x < 1$
$= \frac{2}{x}$, if $1 \leq x < 4$
$= 3$, if $x \geq 4$.

21. Find all critical numbers of the function $f(x) = 2x^4 - 4x^2 + 3$ and classify each critical point as a relative maximum, a relative minimum or neither.

(1 × 10 = 10 marks)